

Geometry of Recursive Knots:

Iterated Cables by Frame Offsets, Their Thickness and Ropelength

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Abstract

We study a concrete family of *recursive knots*: starting from a round circle K_0 , the curve K_d is the (p_d, q_d) torus pattern placed on the tube of radius r_d around K_{d-1} , using a rotation-minimising (Bishop) frame closed by a linear twist. The question of the research line is how much rope (length divided by thickness) such nested knots need. We prove elementary facts for this family: K_d is an embedded smooth closed curve whenever $r_d < \text{thick}(K_{d-1})$ (Lemma 3.1); the closed Bishop frame has linking number $\text{Wr } K_{d-1} - \alpha_d/2\pi$ with its core, which identifies K_d as an explicit cable of K_{d-1} (Lemma 3.2); $p_d L(K_{d-1})(1 - r_d \kappa_{d-1}) \leq L(K_d) \leq p_d L(K_{d-1})(1 + r_d \kappa_{d-1}) + r_d(2\pi q_d + p_d |\alpha_d|)$ (Lemma 3.3); and, for $p_d = 2$, $\text{thick}(K_d) \leq r_d$, so that $\text{Rop}(K_d) \geq L(K_d)/r_d$ inside the family (Proposition 3.5). For the lower bound $\text{thick}(K_d) \geq \min(\text{thick } K_{d-1} - r_d, c r_d \sin(\pi/p_d))$ we prove, for $p_d = 2$ and $r_d \leq \text{thick}(K_{d-1})/2$, that every pair of points of K_d on different strands is at distance $\geq \min(2(\text{thick } K_{d-1} - r_d), 2c_1 r_d)$ with an explicit c_1 depending on $f = r_d / \text{thick } K_{d-1}$ and on the pitch of the strands; in the default family (2,3) this gives $c_1 \geq 0.6117 > 1/2$ at every depth (Lemma 3.7). The curvature of K_d and the same-strand pairs are controlled only under an additional bound on $|K'_{d-1}|$ and $|\kappa'_{d-1}|$ (Propositions 3.8–3.9), and we show that without such a bound no constant $c > 0$ works (Remark 3.11); the remaining case—same-strand pairs at $f = 1/2$, and uniformity in the depth—is stated as a hypothesis (H_c) on the family and checked numerically. Under (H_c) the ropelength bound satisfies a linear recursion whose solution grows geometrically, $\text{Rop}_d \leq 2\pi\Lambda^d$ with an explicit Λ . Numerically, explicit polygons with up to 8192 vertices give $L/\tau = 38.4, 155.6$ and 622.7 at depths 1, 2, 3 for the default pattern (2,3) and offset fraction $f = 1/2$, stable to 0.03% under doubling of the resolution. These are numerically verified upper bounds for explicit polygons, not certified smooth thicknesses; no stationarity, optimality or recursive flow law is claimed.

1 Introduction and status

The research line. Line RTK001 (“Geometry of recursive knots”) asks how nested knots—knots tied on the tube of other knots, recursively—can be represented as tubes without self-intersection, and how much length that costs. Its public record states one finding (a feasible upper bound at depth one) and an explicit scope restriction: neither stationarity nor optimality of any configuration is certified, and no general recursive flow law has been established. No manuscript of the line existed before this draft; the family below was fixed from the record. Version 0.2 incorporates one round of internal review (30/09/2026): the thickness hypothesis of v0.1 was false as stated for an arbitrary base curve (Remark 3.11) and is now a statement about the family, partially proved (Theorem 3.10); the review also corrected the description of the uncovered case and of the numerical criteria (Section 4).

What is done here. We (i) fix a definite family of recursive knots (Section 2); (ii) prove what is elementary about it—embeddedness, the framing it uses, two-sided length bounds, an upper bound on the thickness—and prove part of the thickness lower bound, isolating in one hypothesis on the family what is not proved (Section 3); (iii) compute explicit polygonal configurations at depths 1 to 3, their polygonal thickness and ropelength, with a resolution check (Section 4); (iv) tabulate every claim with its status (Section 5). Lower bounds for general knots are only quoted from the literature: the crossing number of iterated cables is not known in general, so the only lower bound we derive is internal to the family (Proposition 3.5).

2 Setting

2.1 Thickness and ropelength

Let $K \subset \mathbb{R}^3$ be a closed $C^{1,1}$ curve of length $L(K)$. Its *thickness* $\text{thick}(K)$ is the supremum of ρ such that the open tube of radius ρ about K is embedded (normal injectivity radius). Two equivalent descriptions are classical. Litherland, Simon, Durumeric and Rawdon [9] show, for C^2 curves (extended to the $C^{1,1}$ case in [3]; all curves K_d below are C^∞),

$$\text{thick}(K) = \min(\min\text{Rad}(K), \tfrac{1}{2} \text{dcsd}(K)), \quad (1)$$

where $\min\text{Rad}$ is the minimal radius of curvature and dcsd the *doubly critical self distance*, the minimal distance between pairs $x \neq y$ of K whose chord is perpendicular to K at both ends. Gonzalez and Maddocks [7] show $\text{thick}(K) = \inf_{x \neq y} \rho_{pt}(x, y)$, with $\rho_{pt}(x, y) = |x - y| / (2 \sin \angle(T_x, y - x))$ the radius of the circle through y tangent to K at x . The *ropelength* is $\text{Rop}(K) = L(K) / \text{thick}(K)$; it is scale invariant, $\text{Rop}(K) \geq 2\pi$ for every closed curve (Fenchel's theorem [5], total curvature $\geq 2\pi$, together with $\kappa \leq 1 / \text{thick}$), with equality only for the circle, and $\text{thick}(K) \leq \min\text{Rad}(K)$ always.

2.2 Frames, holonomy and the closing twist

Let K be a closed C^2 regular curve parametrised on $[0, 2\pi)$ (not necessarily by arclength), with unit tangent T and curvature κ . A *rotation-minimising* or Bishop frame [1] is an orthonormal frame (T, N_0, B_0) with $N'_0 = -\kappa_1 |K'| T$, $B'_0 = -\kappa_2 |K'| T$, $\kappa_1^2 + \kappa_2^2 = \kappa^2$; it is unique up to a constant rotation in the normal plane and is in general *not* periodic: after one turn the transported normal is rotated by a holonomy angle $\alpha \in (-\pi, \pi]$. We close it by the linear twist $\omega(t) = -\alpha t / 2\pi$, i.e. $N = \cos \omega N_0 + \sin \omega B_0$, $B = T \times N$. The closed frame satisfies

$$N' = -\tilde{\kappa}_1 |K'| T + \omega' B, \quad B' = -\tilde{\kappa}_2 |K'| T - \omega' N, \quad |\omega'| = \frac{|\alpha|}{2\pi} \leq \frac{1}{2}, \quad (2)$$

with the rotated curvature components $\tilde{\kappa}_1 = \kappa_1 \cos \omega + \kappa_2 \sin \omega$, $\tilde{\kappa}_2 = -\kappa_1 \sin \omega + \kappa_2 \cos \omega$ (same norm κ); below we write κ_1, κ_2 for whichever frame is in use.

2.3 The family of recursive knots

Definition 2.1 (iterated cable by frame offsets). Let K_0 be the circle of radius $R_0 = 1$. For $d \geq 1$ choose coprime integers $p_d \geq 2$, $q_d \geq 1$, a radius $r_d > 0$ and a phase φ_d . Reparametrise K_{d-1} to period 2π by the linear change $t \mapsto t/p_{d-1}$ (so that K_{d-1} is traversed at parameter speed $|K'_{d-1}|$, not at unit speed), let (N_{d-1}, B_{d-1}) be its closed Bishop frame, and set for $t \in [0, 2\pi p_d)$

$$K_d(t) = K_{d-1}(t \bmod 2\pi) + r_d [\cos \theta_d(t) N_{d-1}(t) + \sin \theta_d(t) B_{d-1}(t)], \quad \theta_d(t) = \frac{q_d}{p_d} t + \varphi_d. \quad (3)$$

The default is $(p_d, q_d) = (2, 3)$, $\varphi_d = 0$ at every depth; r_d is a fraction f of the thickness of K_{d-1} (the guaranteed lower bound ρ_{d-1} in the theory, the measured polygonal value in the experiments).

K_1 is the (p_1, q_1) torus knot on the round torus of radii $(1, r_1)$; the knot type of K_d for $d \geq 2$ depends on the framing, which Lemma 3.2 makes explicit. The pattern angle advances in the parameter t , not in arclength; since the parametrisation of K_{d-1} is the one inherited from the construction, the *pitch* of the strands relative to the base (the quantity h of Section 3) is a property of the family, not a free parameter.

2.4 Polygonal measurements

Numerically a curve is a closed polygon $P = (x_0, \dots, x_{N-1})$. We use $L(P) = \sum |x_{i+1} - x_i|$; $\min\text{Rad}(P)$, the minimum circumradius of consecutive triples; $\text{dcsd}(P)$, the minimal distance $|x_i - x_j|$ over *doubly critical vertex pairs*, i.e. pairs where $|x_i - x_j|$ is a local minimum in i and in j (the diagonal $D_{ii} = 0$ is kept in the neighbourhood test, which excludes near-diagonal pairs); and the proxy

$$\tau(P) = \min(\min\text{Rad}(P), \tfrac{1}{2} \text{dcsd}(P)), \quad \text{Rop}(P) = L(P) / \tau(P). \quad (4)$$

The local-minimum test could miss doubly critical pairs of saddle type; as a safeguard we also compute $\tau_{pt}(P) = \min_{i \neq j} |x_j - x_i|^2 / (2|T_i \times (x_j - x_i)|)$ with central-difference tangents T_i , which uses no criticality

test. Neither τ nor τ_{pt} is a certified thickness of a smooth curve (for the thickness of polygons themselves see [11]); they are the quantities of (1) evaluated on an explicit polygon, and their stability under doubling of N is the main evidence offered that they approximate the smooth values. Vertex pairs may overestimate the polygon–polygon distance; the minimal segment–segment distance over pairs of edges with cyclic index separation $> N/10$, divided by the vertex dcscd, is 0.99994, 1.00000 ($d = 1, f = 0.25, 0.5$) and 0.99993 ($d = 2, f = 0.5$), and near the critical pair the ratio is ≥ 0.99998 in every level examined (`experiments/aux_checks.py`). Writhe is the Gauss double sum over pairs of edges (midpoint rule); its accuracy is discussed in Section 4.

3 Results

Throughout, $K = K_{d-1}$ is a closed C^∞ regular curve of thickness $\tau = \text{thick}(K) > 0$, hence of maximal curvature $\kappa_{\max} \leq 1/\tau$, carrying the parametrisation of period 2π inherited from Definition 2.1, with speed $v = |K'| \in [v_{\min}, v_{\max}]$; K_d is built by (3) with $r = r_d, p = p_d, q = q_d$, and $f = r/\tau$.

Lemma 3.1 (embedding). *[proved here] If $r < \tau$ then K_d is a regular C^∞ closed curve, injective on $[0, 2\pi p)$, contained in the torus $\partial\{x : \text{dist}(x, K) < r\}$.*

Proof. Smoothness: the Bishop frame of a C^∞ curve is C^∞ (it solves a linear ODE with smooth coefficients), and (3) is a smooth expression in it; periodicity of K_d in t with period $2\pi p$ holds because $\theta_d(t + 2\pi p) = \theta_d(t) + 2\pi q$ and the closed frame has period 2π . Regularity: differentiating (3) with (2),

$$K'_d = |K'| (1 - r(\kappa_1 \cos \theta + \kappa_2 \sin \theta)) T + r(\theta' + \omega')(-\sin \theta N + \cos \theta B), \quad (5)$$

and $|\kappa_1 \cos \theta + \kappa_2 \sin \theta| \leq \kappa \leq 1/\tau$, so the tangential coefficient is at least $|K'|(1 - r/\tau) > 0$. Injectivity: let $t_1 \neq t_2$ in $[0, 2\pi p)$. If $t_1 \not\equiv t_2 \pmod{2\pi}$, the two points lie in the normal discs of radius $r < \tau$ at distinct points of K ; by definition of thickness these discs are disjoint, so the points differ. If $t_1 \equiv t_2 \pmod{2\pi}$, then $t_1 - t_2 = 2\pi k$ with $0 < |k| < p$ and $\theta_d(t_1) - \theta_d(t_2) = 2\pi qk/p \notin 2\pi\mathbb{Z}$ because $\gcd(p, q) = 1$; the two points lie in the same normal disc at the same radius $r > 0$ but at different angles, hence differ. Containment in the torus is immediate from (3). $\square$

The constant $1/2$ in $r \leq \tau/2$, used from Lemma 3.3 on, is what the estimates below need; embeddedness only needs $r < \tau$.

Lemma 3.2 (the framing used, and the knot type). *[proved here, from classical results] Let $\alpha \in (-\pi, \pi]$ be the holonomy of the Bishop frame of K . Then the closed frame (N, B) is a framing of K with linking number $n := \text{Lk} = \text{Wr } K - \alpha/2\pi \in \mathbb{Z}$; in particular $\alpha \equiv 2\pi \text{Wr } K \pmod{2\pi}$, and $n = \text{round}(\text{Wr } K)$ whenever $\text{Wr } K \notin \mathbb{Z} + \frac{1}{2}$ (if $\text{Wr } K \in \mathbb{Z} + \frac{1}{2}$ then $\alpha = \pi$ and $n = \text{Wr } K - \frac{1}{2}$). With the convention that a (p, q') -cable of K winds p times along K and q' times around it relative to a framing of linking number 0 (the Seifert framing), K_d is the $(p, q + pn)$ -cable of K [8]. In particular K_1 is the torus knot $T(p_1, q_1)$.*

Proof. By the Călugăreanu–White–Fuller formula [4, 6, 12], $\text{Lk} = \text{Tw} + \text{Wr}$ for any framing. The Bishop frame has zero twist density; the closing rotation ω adds total twist $\text{Tw} = \frac{1}{2\pi} \int \omega' dt = -\alpha/2\pi \in [-\frac{1}{2}, \frac{1}{2})$. Hence $\text{Lk} = \text{Wr} - \alpha/2\pi$, and since $\text{Lk} \in \mathbb{Z}$ the congruence and the rounding rule follow. The pattern (3) winds q times around the meridian relative to the frame (N, B) while going p times along K ; relative to a framing of linking number 0 it winds $q + pn$ times, which is the stated cable. For K_0 a planar circle, $\text{Wr} = 0 = \alpha$ and K_1 is the standard torus knot. $\square$

Lemma 3.3 (length, two-sided). *[proved here] If $r \leq \tau/2$ then*

$$p L(K) (1 - r\kappa_{\max}) \leq L(K_d) \leq p L(K) (1 + r\kappa_{\max}) + r(2\pi q + p|\alpha|) \leq p L(K) (1 + r/\tau) + r(2\pi q + \pi p). \quad (6)$$

Proof. From (5) and the triangle inequality, $|K'_d| \leq |K'| (1 + r\kappa) + r(|\theta'| + |\omega'|) = |K'| (1 + r\kappa) + r(q/p + |\alpha|/2\pi)$. Integrate over $[0, 2\pi p)$: the first term gives $p \int_0^{2\pi} |K'| (1 + r\kappa) dt \leq p L(K) (1 + r\kappa_{\max})$, the second $r(2\pi q + p|\alpha|)$. Then use $\kappa_{\max} \leq 1/\tau$ and $|\alpha| \leq \pi$. For the lower bound, $|K'_d|$ is at least its tangential component $|K'| (1 - r\kappa) \geq |K'| (1 - r\kappa_{\max})$; integrate. $\square$

Remark 3.4. The term $p|\alpha|r$ from the closing twist is absent from a naive count “ p copies plus q meridians”; it is at most πpr and is needed for the proof via (5) because the Bishop frame of a non-planar K does not close. Numerically the upper bound without it also holds in all 36 measured levels (minimal ratio bound/ L equal to 1.255), so the term is a proof device, not a numerical necessity.

Proposition 3.5 (upper bound on the thickness and a lower bound on the ropelength, $p = 2$). *[proved here] Let $p = 2$ (q odd) and $r < \tau$. Then $\text{thick}(K_d) \leq r$, hence $\text{Rop}(K_d) \geq L(K_d)/r \geq pL(K)(1 - r\kappa_{\max})/r$. In the family with $r_d = f \text{thick}(K_{d-1})$ and $f \leq 1/2$ this gives $\text{Rop}(K_d) \geq \frac{p(1-f)}{f} \text{Rop}(K_{d-1})$, so $\text{Rop}(K_d) \geq 2\pi(p(1-f)/f)^d$.*

Proof. The two points $K_d(t)$ and $K_d(t + 2\pi)$ lie in the same normal disc of K at antipodal angles (θ and $\theta + \pi q$, q odd), so their chord is $-2r(\cos \theta N + \sin \theta B)$, orthogonal to T and to $-\sin \theta N + \cos \theta B$, hence by (5) orthogonal to K'_d at both ends: the pair is doubly critical at distance $2r$, so $\text{dcsd}(K_d) \leq 2r$ and $\text{thick}(K_d) \leq r$ by (1) (equivalently, $\rho_{pt} = r$ for this pair). The rest is Lemma 3.3 with $\kappa_{\max} \leq 1/\tau$, $r = f\tau$ and $\text{Rop}(K_{d-1}) = L(K_{d-1})/\tau$. $\square$

For the default family at $f = 1/2$ the internal lower bound is $2\pi \cdot 2^d$, i.e. 12.6, 25.1, 50.3 at $d = 1, 2, 3$, against the measured 38.4, 155.6, 622.7; with the measured $L(K_d)$ the bound $L(K_d)/r_d$ is attained whenever $\text{thick}(K_d) = r_d$ (Section 4). The lower bound of (6) holds in all 36 measured levels (36 of 36).

Thickness of K_d : what is proved and what is not. Write $s = 2r \sin(\pi/p)$ for the chord between adjacent strands in a normal disc of K (the p points $K_d(t + 2\pi k)$, $k = 0, \dots, p-1$, sit at the p equally spaced angles $\theta + 2\pi qk/p$, since $\gcd(p, q) = 1$).

Proposition 3.6 (two of the three cases). *[proved here] If $r \leq \tau/2$: (a) two points of K_d in the same normal disc of K are at distance $\geq s$; (b) two points of K_d whose base points on K are at distance $\geq 2\tau$ are at distance $\geq 2(\tau - r)$.*

Proof. (a) They are at the same radius r and at angles differing by $2\pi qk/p$, $0 < k < p$, so their distance is $2r|\sin(\pi qk/p)| \geq 2r \sin(\pi/p) = s$ because $qk \not\equiv 0 \pmod{p}$. (b) Triangle inequality: $|K_d(t_1) - K_d(t_2)| \geq |K(t_1) - K(t_2)| - 2r$. $\square$

Not covered by Proposition 3.6 are the pairs with $0 < |K(t_1) - K(t_2)| < 2\tau$, and the curvature of K_d . These pairs include (i) nearby base points (the local helix geometry of p strands on a bent tube) and (ii) base points on different arcs of K at distance $< 2\tau$, e.g. facing each other across the hole of a torus. Case (ii) is not a curiosity: at $f = 1/2$, $d = 1$ the doubly critical vertex pair that realises dcsd has base points 103.7° apart on K_0 , at Euclidean distance $1.573 < 2\tau_0 = 2$ (`aux_checks.py`), while for $f \leq 0.35$ it is the antipodal pair of one normal disc. For a straight tube the p strands are helices with pitch parameter $h = |K'|p/q$ (arclength of the base per radian of strand rotation) and the antipodal pair ($p = 2$) is a local minimum of the strand distance exactly when $h \geq r$; at depth one the relevant pitch on the inner equator is $(1-r)p/q$, and $(1-r)p/q \geq r$ iff $r \leq p/(p+q) = 0.4$ for (2, 3), which matches the observed $\text{thick}(K_1) = r_1$ for $f \leq 0.35$ and $\text{thick}(K_1) < r_1$ for $f = 0.5$. What follows makes part of this quantitative.

Notation for the local case. Let (T, N_0, B_0) be the (non-periodic) Bishop frame of K and α its holonomy, and set

$$m := \frac{q}{p} - \frac{\alpha}{2\pi}, \quad h_{\min} := \frac{v_{\min}}{m}, \quad \Theta_{\min} := \frac{h_{\min}}{\tau}, \quad \zeta_{\min} := \frac{(1-f)h_{\min}}{r}, \quad f = \frac{r}{\tau} \leq \frac{1}{2}.$$

In the parallel frame the strand of (3) turns at the constant rate m (its angle is $\psi = \theta + \omega = mt + \varphi$), so $h = v/m$ is the pitch parameter above ($|K'|p/q$ when $\alpha = 0$) and $\Theta_{\min}$ is the smallest pitch in units of the base thickness. For two points $X_i = K(t_i) + rU_i$ of K_d with distinct base points, let σ be the length of the *shorter* base arc between $K(t_1)$ and $K(t_2)$, $\delta = \sigma/\tau$, s the base-parameter difference along that arc and $u = m|s|$ the relative turning of the strands along it. The integer $k = (t_2 - t_1 - s)/2\pi \pmod{p}$ is the strand index of X_2 relative to X_1 *along that arc* (for $p = 2$ the two strands are exchanged by one turn around K , so “same strand” is meaningful only relative to a base arc). With $\ell_0(\delta) = 2\tau \sin(\delta/2)$, $E(\delta) = \tau(1 - \cos \delta)$ for $\delta \leq \pi/2$ and $E(\delta) = \tau(1 + \delta - \pi/2)$ for $\delta > \pi/2$, $\beta(\delta) = \delta^2/(4 \sin(\delta/2))$, define

$$\Lambda(\delta) = \ell_0 - \frac{2rE}{\ell_0}, \quad P_1(\delta) = \left[2 \cos\left(\frac{\delta}{2\Theta_{\min}}\right) \sqrt{1 - \beta(\delta)^2} - \frac{\delta^2}{3} \right]_+ \quad (7)$$

(with $P_1 := 0$ if $\delta > \pi\Theta_{\min}$ or $\beta > 1$), and

$$c_1(f, \Theta_{\min}) := \frac{1}{2r} \inf_{0 < \delta < \pi} \max \left\{ \sqrt{\Lambda(\delta)_+^2 + r^2 P_1(\delta)^2}, \ell_0(\delta) - 2r \right\}. \quad (8)$$

All lengths in (8) scale with r , so c_1 depends only on $(f, \Theta_{\min})$; $\ell_0/r - 2$ and $\Lambda/r = (2/f) \sin(\delta/2) - (E/\tau)/\sin(\delta/2)$ decrease in f and P_1 does not depend on f and increases in $\Theta_{\min}$, so c_1 is non-increasing in f and non-decreasing in $\Theta_{\min}$.

Lemma 3.7 (pairs on different strands, $p = 2$). *[proved here] Let $p = 2$, q odd, $r \leq \tau/2$. Any two points X_1, X_2 of K_d with distinct base points and strand index $k = 1$ relative to the shorter base arc satisfy*

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c_1(f, \Theta_{\min})r).$$

Moreover $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$ (its numerical value is 0.6117), hence $c_1 \geq 0.6117 > 1/2$ whenever $f \leq \frac{1}{2}$ and $\Theta_{\min} \geq \frac{2}{3}$. For the default family $(p, q) = (2, 3)$ with $r_d \leq \text{thick}(K_{d-1})/2$ one has $\Theta_{\min} \geq \frac{2}{3}$ at every depth, so every such pair is at distance $\geq \min(2(\tau - r), 1.223r)$.

Proof. Step 1 (chord of a thick curve). Let $x = K(0)$ and $g(\sigma) = |K(\sigma) - x|$ in arclength. A local minimum y of g in $(0, L)$ has $(y - x) \perp T(y)$, hence $\rho_{pt}(y, x) = |x - y|/2 \geq \tau$, so $|x - y| \geq 2\tau$. Consequently $\{g < 2\tau\}$ consists of an initial interval $[0, a)$ and a final one $(L - b, L)$ on which g is monotone (a non-monotone stretch below 2τ would contain an interior local minimum below 2τ), and on them $g' = \cos \angle(T, K - x) \geq \sqrt{1 - g^2/4\tau^2}$ a.e. (from $\rho_{pt}(K(\sigma), x) \geq \tau$), which integrates to $g(\sigma) \geq 2\tau \sin(\sigma/2\tau)$ and shows $a, b \leq \pi\tau$ (this is the classical chord-arc inequality for thick curves, cf. [3, 7]). Hence for base points with shorter arc $\sigma = \tau\delta \leq L/2$: either $\delta \geq \pi$ and $|x - y| \geq 2\tau$ (then Proposition 3.6(b) gives $|X_2 - X_1| \geq 2(\tau - r)$), or $\delta < \pi$ and $\ell := |x - y| \geq \ell_0(\delta)$.

Step 2 (elementary estimates, $\delta < \pi$). Let $y = K(t_2)$, $e = (y - x)/\ell$ and $\bar{T} = \sigma^{-1} \int vT dt$ over the arc, so $e = \bar{T}/|\bar{T}|$, $|\bar{T}| = \ell/\sigma$. Since $|dT/d\sigma| = \kappa \leq 1/\tau$: (E2) $|\bar{T} - T(x)| \leq \delta/2$; (E3) for $w \perp T(x)$, $|w \cdot e| = |w \cdot \Pi_x e| \leq |w| |\bar{T} - T(x)| \sigma/\ell \leq |w| \beta(\delta)$, where Π_x is the projection onto the normal plane at x ; (E4) for unit $w \perp T(x)$, $|w \cdot (y - x)| = |\int v w \cdot \Pi_x T dt| \leq \int_0^\sigma \sin \min(\sigma'/\tau, \pi/2) d\sigma' = E(\delta)$, and likewise for $w \perp T(y)$; (E5) if $\tilde{U}_2$ denotes the vector with the same parallel-frame angle as U_2 but in the normal plane at x , then $U_2 - \tilde{U}_2 = -\int \kappa_\perp vT dt$ with $|\kappa_\perp| \leq \kappa$ (differentiate $\cos \psi_2 N_0 + \sin \psi_2 B_0$ along the arc), so $|U_2 - \tilde{U}_2| \leq \delta$ and its component orthogonal to e is at most $\int \kappa v |T - \bar{T}| dt \leq \int_0^\sigma \frac{\sigma'^2 + (\sigma - \sigma')^2}{2\sigma\tau^2} d\sigma' = \delta^2/3$; (E6) $\tilde{U}_2 - U_1$ lies in the normal plane at x and has length $2|\sin(\Delta\psi/2)|$, where $\Delta\psi = ms + 2\pi qk/p$ is the relative angle of the strands after parallel transport along the arc (the holonomy terms cancel because $N_0(t + 2\pi)$ is $N_0(t)$ rotated by α and m contains $-\alpha/2\pi$), i.e. $|\sin(\Delta\psi/2)| = |\cos(u/2)|$ for $k = 1$ and $p = 2$.

Step 3 (decomposition). Write $X_2 - X_1 = (y - x) + r(U_2 - U_1) = Ae + Q$ with $Q \perp e$. By (E4), $A = \ell + r(U_2 - U_1) \cdot e \geq \ell - 2rE/\ell \geq \Lambda(\delta)$, since $\ell \mapsto \ell - 2rE/\ell$ is increasing and $\ell \geq \ell_0$. By (E3), (E5), (E6), $|Q| = r|(U_2 - U_1)_\perp| \geq r[2|\cos(u/2)|\sqrt{1 - \beta^2} - \delta^2/3]_+$. Finally $u = m|s| \leq m\sigma/v_{\min} = \delta/\Theta_{\min}$ and $|\cos(u/2)|$ is decreasing on $[0, \pi]$, so $|\cos(u/2)| \geq \cos(\delta/2\Theta_{\min})$ when $\delta \leq \pi\Theta_{\min}$; together with the trivial bound $|X_2 - X_1| \geq \ell - 2r \geq \ell_0 - 2r$ this gives $|X_2 - X_1| \geq 2c_1(f, \Theta_{\min})r$ for every pair with $\delta < \pi$.

Step 4 ($c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$). Put $r = 1$, $\tau = 2$, $\Theta_{\min} = 2/3$; we show that the maximum in (8) is ≥ 1 for every $\delta \in (0, \pi)$. For $\delta \in [\pi/3, \pi/2]$, $\Lambda = 2(\tau - r) \sin(\delta/2) = 2 \sin(\delta/2) \geq 1$. For $\delta \in [\pi/2, \pi]$, with $\varsigma = \sin(\delta/2) \in [1/\sqrt{2}, 1]$, $\Lambda = 4\varsigma - (1 + 2 \arcsin \varsigma - \pi/2)/\varsigma$, and $\Lambda \geq 1$ amounts to $\phi(\varsigma) := 4\varsigma^2 - \varsigma - 1 + \pi/2 - 2 \arcsin \varsigma \geq 0$; $\phi'' = 8 - 2\varsigma(1 - \varsigma^2)^{-3/2}$ changes sign once on the interval, so ϕ' is increasing then decreasing, and since $\phi'(1/\sqrt{2}) > 0$ and $\phi'(1^-) = -\infty$, ϕ increases then decreases; its minimum is at an endpoint, $\phi(1/\sqrt{2}) = 0.29$, $\phi(1) = 0.43$. For $\delta \in (0, \pi/3)$ we need $4 \sin^2(\delta/2) + P_1^2 \geq 1$. On $[0.6, \pi/3]$: $4 \sin^2(\delta/2) \geq 4 \sin^2(0.3) \geq 0.349$, $\cos(3\delta/4) \geq \cos(\pi/4)$, $\beta \leq \beta(\pi/3) \leq 0.548$ (β is increasing), $\delta^2/3 \leq 0.366$, so $P_1 \geq 2 \cdot 0.7071 \cdot 0.836 - 0.366 \geq 0.816$ and $P_1^2 \geq 0.666$. On $(0, 0.6]$: $\cos(3\delta/4) \geq \cos(0.45) \geq 0.900$, $\beta \leq \beta(0.6) \leq 0.305$, $\delta^2/3 \leq 0.12$, so $P_1 \geq 2 \cdot 0.900 \cdot 0.952 - 0.12 \geq 1.59$. Hence $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$; the grid value 0.6117 (attained near $\delta = 1.23$) is from `theory/check_Hc.py`.

Step 5 (the family). $\Theta_{\min} = v_{\min}/(m\tau)$. By Proposition 3.5, $\tau_{d-1} = \text{thick}(K_{d-1}) \leq r_{d-1} \leq \tau_{d-2}/2$, so $\tau_{d-1} \leq 2^{-(d-1)}$. After the reparametrisation $t \mapsto t/p$, (5) gives $v_{d,\min} \geq p(1 - f)v_{d-1,\min} \geq v_{d-1,\min}$ for $p = 2$, $f \leq 1/2$, so $v_{\min} \geq v_0 = 1$ at every depth; and $m \leq q/p + 1/2 = 2$. Hence $\Theta_{\min} \geq 2^{d-2} \geq 1$ for $d \geq 2$, while for $d = 1$ ($\alpha = 0$, $m = 3/2$, $v = \tau = 1$) $\Theta_{\min} = 2/3$. Monotonicity of c_1 concludes. $\square$

Proposition 3.8 (curvature of K_d). *[proved here, under (H₃)] Assume in addition (H₃): $\sup |v'| \leq V_1$ and $\sup \sqrt{\kappa_1'^2 + \kappa_2'^2} \leq K_1$ for K (derivatives in t , parallel-frame curvatures). Then, with $g(\varsigma) = \varsigma(\varsigma^2 + 2)/(\varsigma^2 + 1)^{3/2}$ and $G(\varsigma_{\min}) = \sup_{\varsigma \geq \varsigma_{\min}} g \leq g(\sqrt{2}) = 1.0887$,*

$$\kappa_{K_d} \leq \bar{\kappa}_d := \frac{G(\varsigma_{\min})}{\tau - r} + \frac{1}{r(1 + \varsigma_{\min}^2)} + \frac{rm[V_1(1 + f) + r v_{\max} K_1]}{(v_{\min}^2(1 - f)^2 + r^2 m^2)^{3/2}}, \quad \min \text{Rad}(K_d) \geq 1/\bar{\kappa}_d. \quad (9)$$

Proof. Let $U = \cos \psi N_0 + \sin \psi B_0$, $W = T \times U$, $\kappa_\perp = \kappa_1 \cos \psi + \kappa_2 \sin \psi$, $\kappa_W = -\kappa_1 \sin \psi + \kappa_2 \cos \psi$ ($\kappa_\perp^2 + \kappa_W^2 = \kappa^2$), $x = r\kappa_\perp \in [-f, f]$. Differentiating (5) with $U' = mW - \kappa_\perp vT$, $W' = -mU - \kappa_W vT$, $T' = v(\kappa_\perp U + \kappa_W W)$:

$$\begin{aligned} K_d' &= c_T T + c_W W, & K_d'' &= aT + b_U U + b_W W, \\ c_T &= v(1-x), & c_W &= rm, & b_U &= v^2(1-x)\kappa_\perp - rm^2, & b_W &= v^2(1-x)\kappa_W, \end{aligned}$$

$a = a_0 - 2rv m \kappa_W$, $a_0 = v'(1-x) - rv(\kappa_1' \cos \psi + \kappa_2' \sin \psi)$. Since (T, U, W) is orthonormal and right-handed, $|K_d' \times K_d''|^2 = b_U^2 |K_d'|^2 + (c_W a - c_T b_W)^2$. Put $A = v^2(1-x)^2$, $B = r^2 m^2$, so $|K_d'|^2 = A + B$ and $c_W a - c_T b_W = r m a_0 - \kappa_W v(A + 2B)$. By Minkowski, $\kappa_{K_d} \leq \sqrt{v^4(1-x)^2 \kappa_\perp^2 (A+B) + \kappa_W^2 v^2 (A+2B)^2} / (A+B)^{3/2} + (rm^2(A+B)^{1/2} + rm|a_0|) / (A+B)^{3/2}$. In the first term $A(A+B) \leq (A+2B)^2$ and $\kappa_\perp^2 + \kappa_W^2 = \kappa^2$ give $\leq v\kappa(A+2B)/(A+B)^{3/2}$, which is decreasing in $A \geq v^2(1-f)^2$; substituting $A = v^2(1-f)^2$ and $\zeta = v(1-f)/(rm)$ yields $\kappa g(\zeta)/(1-f) \leq G(\zeta_{\min})/(\tau(1-f))$. The second term is at most $1/(r(1+\zeta_{\min}^2)) + rm[V_1(1+f) + rv_{\max} K_1]/(v_{\min}^2(1-f)^2 + r^2 m^2)^{3/2}$. $\square$

Proposition 3.9 (pairs on the same strand, $p = 2$). [*proved here, under (H₃)*] Under the hypotheses of Proposition 3.8, every doubly critical pair of K_d with distinct base points and strand index $k = 0$ relative to the shorter base arc satisfies $|X_2 - X_1| \geq \min(2(\tau - r), 2c_0 r)$, where

$$\begin{aligned} c_0 &:= \frac{1}{2r} \inf \left\{ \max \left(\sqrt{\Lambda_+^2 + r^2 P_0^2}, \ell_0 - 2r \right) : 0 < \delta < \pi, \frac{\delta}{\Theta_{\max}} \leq u \leq \frac{\delta}{\Theta_{\min}}, u \geq m s_0 \right\}, \\ P_0 &= \left[2 \left| \sin \frac{u}{2} \right| \sqrt{1 - \beta^2 - \frac{\delta^2}{3}} \right]_+, \end{aligned}$$

$\Theta_{\max} = v_{\max}/(m\tau)$ and $s_0 = \pi/(\bar{\kappa}_d V_d)$, $V_d = v_{\max}(1+f) + rm \geq \sup |K_d'|$.

Proof. For a doubly critical pair, along the arc of K_d that follows the chosen base arc (of K_d -length $\sigma_d \leq V_d |s|$), $0 = (X_2 - X_1) \cdot T_d(X_1) = \int_0^{\sigma_d} T_d \cdot T_d(X_1) d\sigma \geq \int_0^{\sigma_d} \cos(\bar{\kappa}_d \sigma) d\sigma = \sin(\bar{\kappa}_d \sigma_d)/\bar{\kappa}_d$, which is positive unless $\sigma_d \geq \pi/\bar{\kappa}_d$; hence $|s| \geq s_0$ and $u \geq m s_0$. For $k = 0$, $\Delta\psi = \pm u$ in Step 2 of Lemma 3.7, and Step 3 gives the bound with $|\sin(u/2)|$ in place of $|\cos(u/2)|$; the infimum is over the whole admissible range of u , and the lower end $u \geq \delta/\Theta_{\max}$ (from $\sigma \leq v_{\max}|s|$) now matters because small u is the worst case. $\square$

Theorem 3.10 (what is proved of the thickness lower bound for $p = 2$). [*partially proved*] Let $p = 2$, $f = r/\tau \leq 1/2$ and assume (H₃). Then, by (1), Lemma 3.7 and Propositions 3.8–3.9,

$$\text{thick}(K_d) \geq \min(\tau - r, c r), \quad c = \min \left(c_1(f, \Theta_{\min}), c_0, \frac{1}{r\bar{\kappa}_d} \right).$$

The first constant needs no hypothesis beyond $r \leq \tau/2$ and satisfies $c_1 \geq 0.6117 > 1/2$ whenever $\Theta_{\min} \geq 2/3$, in particular throughout the default family (2, 3) with $r_d \leq \text{thick}(K_{d-1})/2$. At $d = 1$ the base is the round circle, (H₃) holds with $V_1 = K_1 = 0$, and all three constants are unconditional. On the polygons of Section 4 ($N_0 = 256$; V_1, K_1 measured on the base polygon for $d \geq 2$) the constants $(c_1, c_0, 1/(r\bar{\kappa}_d))$ are: $f = 0.25$: (1.00, 1.33, 1.79), (1.00, 1.62, 2.90), (1.00, 1.60, 3.00) at $d = 1, 2, 3$; $f = 0.35$: (0.97, 0.65, 1.02), (1.00, 0.66, 1.59), (1.00, 0.67, 1.84); $f = 0.5$: (0.61, 0.32, 0.56), (0.70, 0.13, 0.49), (0.71, 0.16, 0.85). Hence the bound with $c = 1/2$ is proved for $f \leq 0.35$ at $d = 1$ unconditionally and at $d = 2, 3$ under (H₃) with the measured constants, while for $f = 1/2$ it is proved only for pairs on different strands (and, at $d = 1, 3$, for the curvature); the same-strand case at $f = 1/2$ is proved only with $c = c_0 \leq 0.32$.

Remark 3.11 (no constant works for an arbitrary base). [*proved here*] No constant $c > 0$ depending only on (p, q, f) makes $\text{thick}(K_d) \geq \min(\tau - r, c r \sin(\pi/p))$ true for every base K with $\text{thick}(K) = \tau$, even for smooth K —which is why the hypothesis of v0.1, stated for an arbitrary K , was false (the internal review found two counterexamples; we reproduce both). (i) *Curvature derivative.* If the curvature of K rises from 0 to $1/\tau$ over an arc of length ε (a smoothed junction between a segment and a circle of radius τ , whose thickness tends to τ), then by (5) the tangent direction of the inner strand of K_d turns from $vT + rmW$ to $v(1-f)T + rmW$ over the same arc, so $\kappa_{K_d} \gtrsim rv\kappa'/|K_d'|^2 \rightarrow \infty$ and $\text{thick}(K_d) \rightarrow 0$; the term in (9) involving V_1, K_1 is therefore unavoidable. (ii) *Pitch.* If K is traversed slowly on an arc of length $\gg r$ ($v \rightarrow 0$ there, geometry and τ unchanged), then on that arc the two strands ($p = 2$) are helices of pitch $h = v/m \rightarrow 0$: in the straight-tube model the squared distance between $K_d(t)$ and $K_d(t + 2\pi + u)$ is $4r^2 \cos^2(qu/4) + v^2 u^2$, whose minimum over u is at most $(2\pi v/q)^2$ (the value at $u = 2\pi/q$) and is attained at a doubly critical pair by the screw symmetry, so $\text{thick}(K_d) \leq \pi v/q \rightarrow 0$; in Lemma 3.7 this is

the dependence on $\Theta_{\min}$ ($c_1 \rightarrow 0$ as $\Theta_{\min} \rightarrow 0$). In the recursive family both quantities are controlled: $\Theta_{\min} \geq 2/3$ a priori (Lemma 3.7), and the K''' -term of (9) is damped by the factor $rm/|K'_d| \approx r/h$, but its uniformity in d is not established here.

What remains unproved is therefore a statement about the family, not about an arbitrary curve:

Hypothesis 3.12 (H_c , for the family). [partially proved (Theorem 3.10); the rest verified numerically for $c = 1/2$] Let (K_d) be the family of Definition 2.1 built from the circle with a fixed pattern (p, q) and radii $r_d \leq f \text{thick}(K_{d-1})$, $f \leq 1/2$. There is $c = c(p, q, f) \in (0, 1]$ such that for every $d \geq 1$

$$\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, c r_d \sin(\pi/p)). \quad (10)$$

Status for (2, 3), $f \leq 1/2$. Proved for all d for pairs on different strands with $c = 0.6117$ (Lemma 3.7). The curvature and the same-strand pairs are proved at $d = 1$, and at $d = 2, 3$ under (H_3) with constants measured on the base polygons; with $c = 1/2$ this covers $f \leq 0.35$ at $d \leq 3$ (Theorem 3.10). Open: the same-strand case at $f = 1/2$ ($c_0 = 0.32, 0.13, 0.16$ at $d = 1, 2, 3$; the measured same-strand doubly critical pairs are far from this bound), the a priori curvature constant at $f = 1/2$, $d = 2$ (0.49), and the uniformity in d of (H_3) . Numerically (Section 4) (10) holds with $c = 1/2$ in all 9 default levels with $f \leq 1/2$, margin $\tau_d/\rho_d \geq 1.66$.

On the constant. The natural constant is $c = 1$: $r \sin(\pi/p) = s/2$ is half the strand chord, Proposition 3.6(a) gives exactly this for same-disc pairs, and it is sharp for $p = 2$ on a straight tube. With $c = 1$, however, the inequality $\tau_d \geq \rho_d$ fails on the polygons in 4 of the 15 measured levels with $f \leq 1/2$ (one of the default family, three of the other patterns): $(2, 3)$, $f = 0.5$, $d = 1$ (0.832); $(3, 2)$, $d = 1$ (0.968); $(3, 2)$, $d = 2$ (0.999); $(2, 5)$, $d = 1$ (0.564); and it is attained with equality, to the precision of the data, in every level where $\tau_d = r_d$ (all default levels with $f \leq 0.35$, and $f = 1/2$ at $d = 2, 3$: by Proposition 3.5, $c = 1$ cannot be improved there) as well as in $(3, 2)$, $d = 2, 3$; $(2, 5)$, $d = 2, 3$, where the critical pair is a pair of adjacent strands of one disc. The value $c = 1/2$ used in the criteria of Section 4 is therefore an empirical choice made after a first run with $c = 1$ had failed at $(f = 1/2, d = 1)$, not a prediction; the proved constants of Theorem 3.10 show that $c = 1/2$ is attainable by the present method for different-strand pairs ($c_1 \geq 0.6117$) but not yet for the same-strand case at $f = 1/2$.

Corollary 3.13 (recursion for the ropelength bound). [proved here, conditional on H_c for the family] Fix (p, q) , $f \leq 1/2$ and assume (H_c) with constant c (for all $r_d \leq f \text{thick}(K_{d-1})$). Define $\rho_0 = 1$, $r_d := f \rho_{d-1}$ and $\rho_d := \gamma \rho_{d-1}$ with

$$\gamma = \min(1 - f, c f \sin(\pi/p)), \quad A = p(1 + f) + f(q + \frac{p}{2}). \quad (11)$$

Then $\text{thick}(K_d) \geq \rho_d = \gamma^d$ and $L(K_d) \leq 2\pi A^d$ for all d , hence $\text{Rop}(K_d) \leq 2\pi \Lambda^d$ with $\Lambda = A/\gamma$. For $(2, 3)$ and $f = 1/2$: $A = 5.00$, and $\Lambda = 20.0$ with $c = 1/2$ ($\gamma = 0.250$), $\Lambda = 10.0$ with $c = 1$ ($\gamma = 0.500$).

Proof. Induction on d . If $\text{thick}(K_{d-1}) \geq \rho_{d-1}$ then $r_d = f \rho_{d-1} \leq f \text{thick}(K_{d-1})$, so (H_c) applies and $\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, c r_d \sin(\pi/p)) \geq \min(\rho_{d-1} - f \rho_{d-1}, c f \rho_{d-1} \sin(\pi/p)) = \gamma \rho_{d-1} = \rho_d$. By Lemma 3.3 with $\tau \geq \rho_{d-1}$, $L(K_d) \leq p L(K_{d-1})(1 + f) + f \rho_{d-1}(2\pi q + \pi p)$; since $\rho_{d-1} \leq \text{thick}(K_{d-1}) \leq L(K_{d-1})/2\pi$, this is $\leq L(K_{d-1})[p(1 + f) + f(q + p/2)] = A L(K_{d-1})$, and $L(K_0) = 2\pi$. $\square$

Remark 3.14 (on the constants). Λ is generous: Lemma 3.3 bounds $|K'_d|$ by the sum of the norms of two orthogonal vectors, uses $\kappa_{\max} \leq 1/\tau$ and $|\alpha| \leq \pi$, and (H_c) with $c = 1/2$ is twice weaker than the observed $\tau_d \approx r_d$. Numerically the growth factor per depth is 4.06, 4.00 at $f = 1/2$ (Table 1) against $\Lambda = 20.0$; the conditional bounds $2\pi \Lambda^d$ are 126, 2513, 50265 against the measured 38.4, 155.6, 622.7 (the two chains use different r_d , so the comparison is indicative only). We make no claim that any configuration in the family is stationary or optimal for ropelength, and none that the recursion is a “flow law” for tight recursive knots: it bounds one explicit construction.

Remark 3.15 (lower bounds, quoted). [literature] Buck and Simon [2] prove $\text{Rop}(K) \geq c_0 \text{Cr}(K)^{3/4}$ for a universal $c_0 > 0$, and Cantarella, Kusner and Sullivan [3] prove that ropelength minimisers exist in every knot and link type, are $C^{1,1}$, and satisfy further lower bounds obtained from cone surfaces. To turn these into a lower bound for the knot type of K_d one needs $\text{Cr}(K_d)$; for the iterated cables of Lemma 3.2 the crossing number is not known in general (crossing numbers of satellites are an open problem), so the only lower bound we have is the one internal to the family, Proposition 3.5. For $K_1 = T(2, 3)$ the best known numerical ropelength is about 32.7 (numerical, uncertified; [3, 10]); our depth-one polygon at $f = 1/2$ gives 38.4, i.e. the construction is not close to tight even at depth one.

Conjecture 3.16 (what the numbers suggest). [conjectural] For the default family with $f \leq 1/2$, $\text{thick}(K_d) = r_d$ exactly whenever the antipodal pair in a normal disc is the doubly critical pair (observed for $f \leq 0.35$ at all depths and for $f = 1/2$ at $d \geq 2$; the inequality $\leq$ is Proposition 3.5, the equality is not proved).

Remark 3.17 (growth rate). If $\text{thick}(K_d) = r_d$ for all large d and $\text{minRad}(K_{d-1})$ stays bounded below, then $\text{Rop}_d / \text{Rop}_{d-1} = (L_d / L_{d-1})(\tau_{d-1} / \tau_d) \rightarrow p/f$ by Lemma 3.3 ($L_d / L_{d-1} \rightarrow p$ because $r_d \kappa_{\max}(K_{d-1}) \leq r_d / \text{minRad}(K_{d-1}) \rightarrow 0$). The measured ratios $\text{Rop}_d / \text{Rop}_{d-1}$ at $d = 2, 3$ are 8.01, 8.00 ($f = 0.25$; $p/f = 8.00$), 5.73, 5.71 ($f = 0.35$; 5.71), 4.06, 4.00 ($f = 0.5$; 4.00), 4.03, 4.00 ((2, 5); 4.00) and 6.95, 6.93 ((3, 2), where the critical pair is a pair of adjacent strands and the limit is $p/(f \sin(\pi/p)) = 6.93$). The rate p/f is thus a consequence of the conjecture, not an independent empirical constant (v0.1 presented it as one).

4 Experiments

Criteria. The protocol is in Appendix A. Four criteria are checked: (C1) resolution: Rop and τ change by less than 1% between N_0 and $2N_0$; (C2) $L(K_d) \leq$ the right-hand side of (6) evaluated with the measured $L(K_{d-1})$, $\tau(K_{d-1})$ and α_d ; (C3) $\tau(K_d) \geq \rho_d$ of (10) with $c = 1/2$ and the measured $\tau(K_{d-1})$, for $f \leq 1/2$; (C4) τ_{pt} agrees with τ to within 1%. Which of these were fixed in advance: the plan written before any code (PLAN.md) fixed the resolution check at $d = 1, 2$ without a threshold, the length bound and the form of (10) with an unspecified constant; the 1% thresholds, the extension of (C1) to $d = 3$ and the cross-check (C4) were fixed when the manuscript was written, and the constant $c = 1/2$ of (C3) was fixed *after* a first run with the natural constant $c = 1$ had failed at ($f = 1/2, d = 1$). (C3) is therefore a post hoc check, reported as such.

Results. Table 1 gives all patterns at the finest resolution; Figure 1 shows the default curves and Figure 2 the growth. All four criteria hold: (C1) the maximal relative change is 0.03% for Rop and 0.03% for τ over all 12 chains (per-level values in `results/tables.md`); (C2) holds in 36 of 36 levels; (C3) holds in all 9 default levels with $f \leq 1/2$, with margin $\tau_d / \rho_d \geq 1.66$ (and ≥ 1.13 including the other patterns); with $c = 1$ instead of $1/2$ the margin drops to 0.56 and the inequality fails in 4 of 15 levels, as discussed after Hypothesis 3.12; (C4) the maximal discrepancy is 0.01%. In every level τ is determined by $\text{dcsd}/2$, never by minRad (36 of 36; $\text{minRad}/r_d \geq 1.15$ throughout).

Three further observations. First, for $f \leq 0.35$ the measured $\tau(K_d)$ equals r_d to four digits at every depth (the antipodal strands in a normal disc are the doubly critical pair, consistent with Proposition 3.5 and Conjecture 3.16), so $\text{Rop}_d = L_d / r_d$ and the recursion $\tau_d = f\tau_{d-1}$ is exact; for $f = 1/2$ at depth one $\tau_1 = 0.4158 < r_1 = 0.5$, realised by the pair across the hole of the torus described in Section 3, while at depths 2, 3 again $\tau_d = r_d$. Second, the writhe of the depth-one configuration crosses the half-integer 3.5 inside the grid ($\text{Wr}(K_1) = 3.127, 3.257, 3.518, 3.710$ for $f = 0.25, 0.35, 0.5, 0.6$; the crossing is at $r_1^* = 0.4904$ by bisection), so by Lemma 3.2 the frame linking number is 3 for $f \leq 0.35$ and 4 for $f \geq 0.5$ and the depth-two knot type changes across the grid: K_2 is the (2, 9)-cable of the trefoil for $f = 0.25$ but the (2, 11)-cable for $f = 0.5$. The frame linking number is computed as $\text{rint}(\text{Wr } K_{d-1} - \alpha_d / 2\pi)$; the residual $|\text{Wr } K_{d-1} - \alpha_d / 2\pi - n|$ is a consistency check of Lemma 3.2 and is at most $1.1 \cdot 10^{-3}$ at the fine resolution ($4.5 \cdot 10^{-3}$ at the coarse one) over all levels. Third, the best depth-one value on the grid is $f = 0.5$ ($\text{Rop}_1 = 38.4$), the best depth-three value also ($\text{Rop}_3 = 622.7$); $f = 0.6$ is worse at every depth, but we did not optimise f and make no optimality claim.

Writhe accuracy. The midpoint-rule writhe changes under doubling of N by at most $1 \cdot 10^{-4}$ at depth one of the default family, but by up to 0.031 at depth three of (3, 2) (the double sum converges slowly on the long chains); the knot-type decisions are unaffected (the smallest margin to a half-integer is 0.018 at $f = 1/2, d = 1$, where the change is $\leq 10^{-4}$), but the accuracy claimed for Wr is level-dependent. An exact polygonal writhe (Levitt; Klenin–Langowski) would remove this at the same cost and is listed as a next step.

Curve-shortening probe (uncertified). `experiments/shrink.py` applies 200 explicit discrete curve-shortening steps (each vertex moved a fraction 0.1 of the way to the midpoint of its neighbours) to K_1 and K_2 of the (2, 3), $f = 1/2$ chain at $N_0 = 512$, tracking the scale-invariant L/τ . At depth one L/τ falls from 38.4 to 38.1 (0.8%, still decreasing at the last step); at depth two no step improves it: L/τ rises

monotonically from 155.6 to 156.3. Plain curve shortening is thus not a cheap way to improve the family and gives no evidence about local minimality; details in `results/tables_shrink.md`.

Table 1: All chains at the finest resolution ($N_0 = 1024$; 512 for $(3, 2)$). $r_d = f\tau_{d-1}$; $\rho_d^{(1)} = \min(\tau_{d-1} - r_d, r_d \sin(\pi/p))$ is (10) with the natural constant $c = 1$, so $\tau_d/\rho_d^{(1)} < 1$ marks a failure of $c = 1$ (the ratio for $c = 1/2$ is twice as large when $\tau_{d-1} - r_d$ is not the minimum); slope = $q + pn$ of Lemma 3.2. † : $f = 0.6$ lies outside the hypothesis $r \leq \tau/2$ and is a numerical probe only. Lengths in units of R_0 .

(p, q)	f	d	N	r_d	L	τ	L/τ	$\tau_d/\rho_d^{(1)}$	Wr	slope
(2, 3)	0.25	1	2048	0.2500	13.444	0.2500	53.8	1.000	3.13	3
(2, 3)	0.25	2	4096	0.0625	26.911	0.0625	430.6	1.000	15.25	9
(2, 3)	0.25	3	8192	0.0156	53.822	0.0156	3444.6	1.000	63.50	33
(2, 3)	0.35	1	2048	0.3500	14.271	0.3500	40.8	1.000	3.26	3
(2, 3)	0.35	2	4096	0.1225	28.608	0.1225	233.5	1.000	15.49	9
(2, 3)	0.35	3	8192	0.0429	57.219	0.0429	1334.6	1.000	63.98	33
(2, 3)	0.5	1	2048	0.5000	15.949	0.4158	38.4	0.832	3.52	3
(2, 3)	0.5	2	4096	0.2079	32.349	0.2079	155.6	1.000	17.99	11
(2, 3)	0.5	3	8192	0.1040	64.730	0.1040	622.7	1.000	74.96	39
(2, 3)	0.6 †	1	2048	0.6000	17.307	0.3401	50.9	0.850	3.71	3
(2, 3)	0.6 †	2	4096	0.2041	34.941	0.1367	255.6	1.005	18.36	11
(2, 3)	0.6 †	3	8192	0.0820	69.893	0.0820	852.1	1.500	75.70	39
(3, 2)	0.5	1	1536	0.5000	20.000	0.4191	47.7	0.968	3.90	2
(3, 2)	0.5	2	4608	0.2096	60.087	0.1813	331.4	0.999	39.66	14
(3, 2)	0.5	3	13824	0.0906	180.270	0.0785	2296.5	1.000	362.97	122
(2, 5)	0.5	1	2048	0.5000	20.420	0.2822	72.4	0.564	6.87	5
(2, 5)	0.5	2	4096	0.1411	41.116	0.1411	291.4	1.000	32.66	19
(2, 5)	0.5	3	8192	0.0705	82.273	0.0705	1166.2	1.000	136.30	71

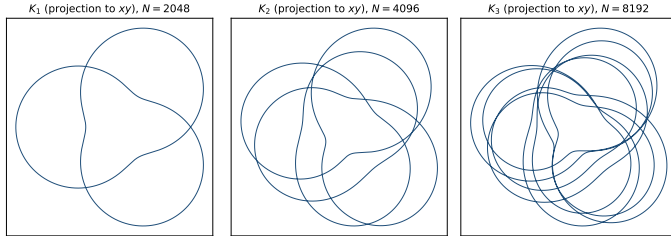


Figure 1: Projections to the xy -plane of K_1, K_2, K_3 for $(2, 3)$, $f = 0.5$.

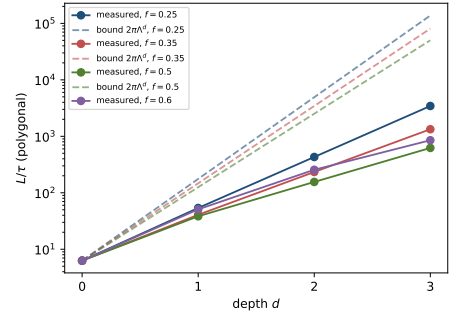


Figure 2: Polygonal L/τ against depth d for the default family (solid) and the conditional bound $2\pi\Lambda^d$ with $c = 1/2$ (dashed, $f \leq 1/2$).

5 Claims and their status

Claim	Status
K_d is embedded, smooth and regular when $r_d < \text{thick}(K_{d-1})$ (Lemma 3.1).	proved here
The closed Bishop frame has $\text{Lk} = \text{Wr} - \alpha/2\pi$; K_d is the $(p, q + pn)$ -cable of K_{d-1} (Lemma 3.2).	proved here from classical results
$pL(K_{d-1})(1 - r\kappa_{\max}) \leq L(K_d) \leq pL(K_{d-1})(1 + r\kappa_{\max}) + r(2\pi q + p \alpha)$ (Lemma 3.3).	proved here
$\text{thick}(K_d) \leq r_d$ and $\text{Rop}(K_d) \geq L(K_d)/r_d \geq 2\pi(p(1-f)/f)^d$ for $p = 2$ (Prop. 3.5).	proved here (lower bound internal to the family)
Same-disc pairs at distance $\geq 2r \sin(\pi/p)$; far pairs at distance $\geq 2(\tau - r)$ (Prop. 3.6).	proved here
Different-strand pairs at distance $\geq \min(2(\tau - r), 2c_1r)$, $c_1 \geq 0.6117$ in the $(2, 3)$ family, $f \leq 1/2$, all d (Lemma 3.7).	proved here
$\text{minRad}(K_d) \geq 1/\bar{\kappa}_d$ and same-strand doubly critical pairs at distance $\geq \min(2(\tau - r), 2c_0r)$ (Props. 3.8–3.9).	proved here under (H_3) ; unconditional at $d = 1$
$\text{thick}(K_d) \geq \min(\tau - r, cr)$ with $c = 1/2$, $p = 2$, $f \leq 0.35$, $d \leq 3$ (Thm. 3.10).	proved at $d = 1$; at $d = 2, 3$ under (H_3) with measured constants
No $c > 0$ depending only on (p, q, f) works for an arbitrary base K (Remark 3.11).	proved here
(H_c) for the family with $c = 1/2$: same-strand case at $f = 1/2$, uniformity in d .	not proved; verified numerically in 9 levels, margin ≥ 1.66 ($c = 1$ fails in 4 of 15 levels)
$\text{Rop}(K_d) \leq 2\pi\Lambda^d$, $\Lambda = A/\gamma$ (Cor. 3.13).	proved conditional on (H_c) for the family
Explicit polygons with $L/\tau = 38.4, 155.6, 622.7$ at $d = 1, 2, 3$ ($f = 0.5$), stable to 0.03%.	computationally verified (polygonal proxy, not certified smooth thickness)
Knot type of K_2 changes between $f = 0.35$ and $f = 0.5$ (writhe of K_1 crosses 3.5 at $r_1^* = 0.4904$).	computationally verified ($\text{Wr}(K_1)$ stable to 10^{-4}), consequence of Lemma 3.2
$\text{Rop} \geq c_0 \text{Cr}^{3/4}$; existence and regularity of tight knots.	literature [2, 3]
$\text{thick}(K_d) = r_d$ for small f (Conj. 3.16); then $\text{Rop}_d / \text{Rop}_{d-1} \rightarrow p/f$.	conjectural; the rate is a consequence, not a new constant
Curve-shortening lowers the polygonal L/τ of K_1 by 0.8% and raises that of K_2 .	uncertified numerical probe; negative at depth two
Stationarity or optimality of any K_d ; a recursive flow law for tight recursive knots.	not claimed

6 Limitations and next steps

Limitations. (1) The thickness lower bound is fully proved only for different-strand pairs ($p = 2$); the curvature of K_d and the same-strand pairs need (H_3) , whose constants are measured on the polygons for $d \geq 2$ and whose uniformity in d is open; at $f = 1/2$ the same-strand constant c_0 falls short of $1/2$. (2) The polygonal τ of (4) uses vertex pairs and a local-minimum test; agreement with τ_{pt} to 0.01%, the segment–segment check of Section 2.4 and stability under doubling of N are evidence, not certification, which would require interval arithmetic on a $C^{1,1}$ interpolant. (3) r_d is a fixed fraction of the measured thickness; nothing is optimised, and the family (pattern angle linear in t , phase $\varphi_d = 0$) is one of many. (4) The conditional bound $2\pi\Lambda^d$ and the measured chain use different r_d . (5) Writhe and holonomy are midpoint-rule sums, accurate to 10^{-4} at depth one but only to 0.031 at depth three of $(3, 2)$; the frame linking number for $f = 0.5$ is decided by a margin of 0.02. (6) The only lower bound specific to the family is Proposition 3.5; the knot-type lower bounds need $\text{Cr}(K_d)$, which is not known. (7) The constants of Theorem 3.10 other than $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$ are infima of explicit functions evaluated on grids with an estimated Lipschitz correction, not with interval arithmetic.

Next steps. (1) At $d = 1$ the uncovered case is an explicit, certifiable computation: the minimal distance between the strands of $T(p, q)$ on the torus of radii $(1, r)$ as a function of r (the critical pair at $f = 1/2$ is the pair across the hole, not a local helix pair); for the same-strand case at $f = 1/2$ and $d \geq 1$, use the exact expression of Proposition 3.8 for $|K'_d \times K''_d|$ with the angle ψ kept, instead of $\kappa_1^2 + \kappa_W^2 = \kappa^2$, to sharpen $\bar{\kappa}_d$ away from the inner side of the tube; and bound V_1, K_1 of K_d in terms of those of K_{d-1} to make (H_3) uniform in d . (2) Replace the polygonal proxy by a certified computation (interval arithmetic on minRad and dcsd of a piecewise-circular interpolant) and the midpoint writhe by the exact polygonal writhe. (3) Replace the curve-shortening probe by a constrained length minimisation at fixed $\tau \geq 1$ with a certified knot-type check, for $d = 1, 2$. (4) Optimise f_d per depth and the pattern angle in arclength rather than in t . (5) Compute the crossing numbers of the specific cables identified in Lemma 3.2 where known, to attach a Buck–Simon lower bound to each explicit configuration.

A Reproducibility and protocol

Protocol. `experiments/recursive_knots.py` builds the polygon K_0 with N_0 vertices, computes the closed Bishop frame by discrete parallel transport (rotation taking T_i to T_{i+1} , Rodrigues formula; initial normal orthogonal to T_0 ; holonomy measured as the signed angle between the transported and the initial normal, removed by the twist $-\alpha i/N$) and produces K_d with $p_d N_{d-1}$ vertices by (3) evaluated at the vertices (no interpolation; the parameter t is the vertex index). At each depth it measures L , $\min\text{Rad}$, dcsd , τ , τ_{pt} , Rop , the writhe, the holonomy and $\text{Lk} = \text{rint}(\text{Wr } K_{d-1} - \alpha_d/2\pi)$ with its residual, and sets $r_{d+1} = f \tau(K_d)$. Grid: $(p, q) = (2, 3)$ with $f \in \{0.25, 0.35, 0.5, 0.6\}$, and $(3, 2)$, $(2, 5)$ with $f = 0.5$; depths 1–3; two resolutions, $N_0 \in \{512, 1024\}$ ($\{256, 512\}$ for $(3, 2)$, whose depth-3 polygon has $27N_0$ vertices). The value $f = 0.6$ lies outside the hypothesis $r \leq \tau/2$ and is included only as a numerical probe. The computation is deterministic (seed 20260930 recorded for bookkeeping only); the reference run took 184s of wall-clock time (3.1 minutes) on shared cores with Python 3.11.15 and NumPy 2.4.6. `experiments/aux_checks.py` computes the critical-pair geometry at depth one, the writhe crossing r_1^* and the segment–segment check (`results/aux_checks.json`); `theory/check_Hc.py` evaluates the constants of Theorem 3.10 on the polygons and checks every intermediate inequality of Lemma 3.7 on all local pairs (`theory/check_Hc_output.txt`, `check_Hc_summary.json`). `python3 experiments/make_numbers.py` writes `manuscript/numbers.tex` and the table body from these JSON files, and `manuscript/build.sh` runs it and `latexmk`. Every number in this text is a macro from `numbers.tex`.

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